

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

github.com/poojakira/dataset-poisoning-detector - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

200

Tests passed

91.20%

Statement coverage

0.55

Cross-class F1 @ 5% poison

0.595

Cross-class F1 @ 10% poison

SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

Verified Evidence

EV-01	Cross-class F1 @ 20% poison: 0.6975	EV-02	Old top-1 spectral F1: 0.08 / 0.23 / 0.3675
EV-03	Feature-space ensemble F1: 0.0841 / 0.1421 / 0.2331	EV-04	200 tests passed, 0 skipped.
EV-05	91.20% statement coverage; CI gate is 90%.	EV-06	Ruff check/format and Pyright pass on the final branch; GitHub currently reports 0 open Code Scanning, Dependabot, and secret-scanning alerts.
EV-07	Committed results/spectral_benchmark.json records cross-class centroid F1:	EV-08	0.55 at 5% contamination
EV-09	0.595 at 10%	EV-10	0.6975 at 20%

Method and Control Path

- 01: Validate incoming feature batches and streaming samples.
- 02: Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.
- 03: Combine detector signals for screening and quarantine workflows.
- 04: Maintain streaming state and deduplication metadata.
- 05: Benchmark label-flip behavior with committed reproducible artifacts.

Limitations and Residual Risk

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.

Reproducibility

Expected current evidence: 200 passed, 91.20% coverage.

Evidence Provenance

- Claim discipline**
Metrics come from the repository's verified metrics and research brief.
- Scope discipline**
Benchmarks are reported only at their documented fixture, dataset, or test scope.
- Reproduction**
PDF and PNG are generated locally from committed evidence without paid services.